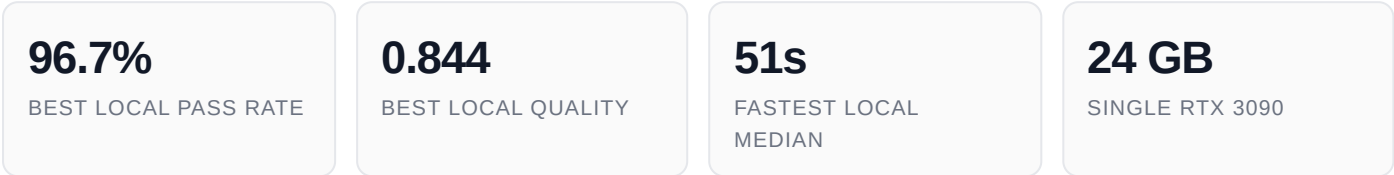


Sovereign Stack Benchmark

Local 30B-class models on a single consumer GPU, measured against hosted frontier models on the same agentic coding and operations tasks — under two different agent harnesses.

Independent single-operator evaluation · 2026-08-12 · 7 models × 15 tasks × 2 harnesses = 210 graded runs



local (self-hosted) frontier (hosted API)

How this was measured

Every model runs the same 15 tasks twice — once driven by each of two agent harnesses — inside an isolated working copy, with no access to the grading tests as an oracle. The agent is told to run `pytest` and iterate until green. Each run is scored on two independent axes:

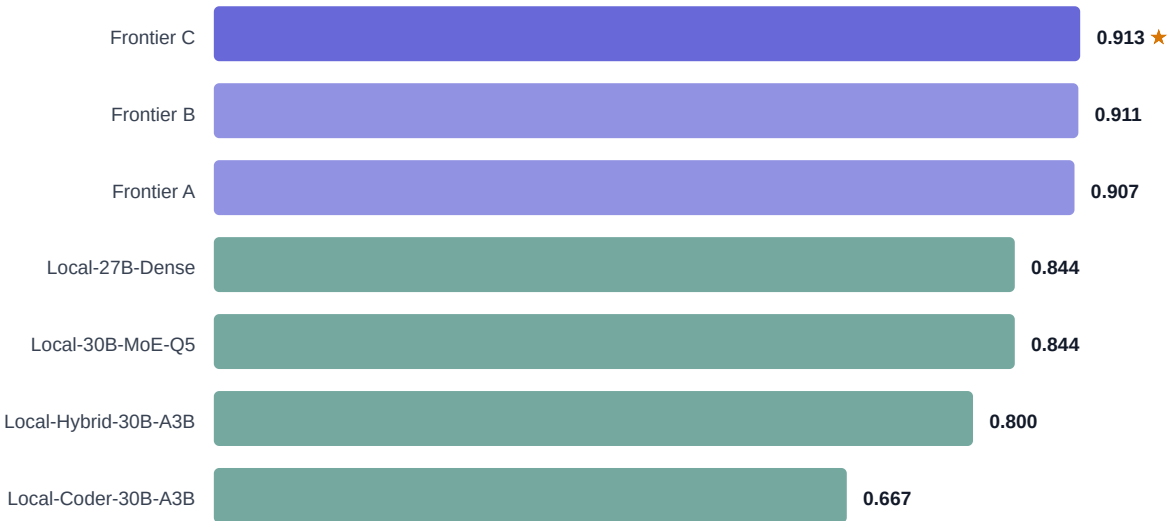
Pass rate is the mean per-test pass fraction of the hidden suite, so partial credit is real: fixing one of two planted bugs scores 0.5. **Quality** is a 0–1 LLM-as-judge score from an independent frontier model, rating the finished solution for correctness, edge-case handling and craft — a fragile pass scores below a clean one. **Latency** is wall-clock time to a finished task, including every tool call and retry the agent made, not token throughput.

The task battery was curated with Fable 5 and mirrors my day-to-day agentic work: AIOps alert correlation, NetDevOps intent-to-ACL compilation and config-drift planning, HPC backfill scheduling, inference KV-budget batching, RAG pipeline debugging and tool-calling loops — alongside held-out EvalPlus coding problems, cross-module refactors and long-context needle retrieval.

Leaderboard

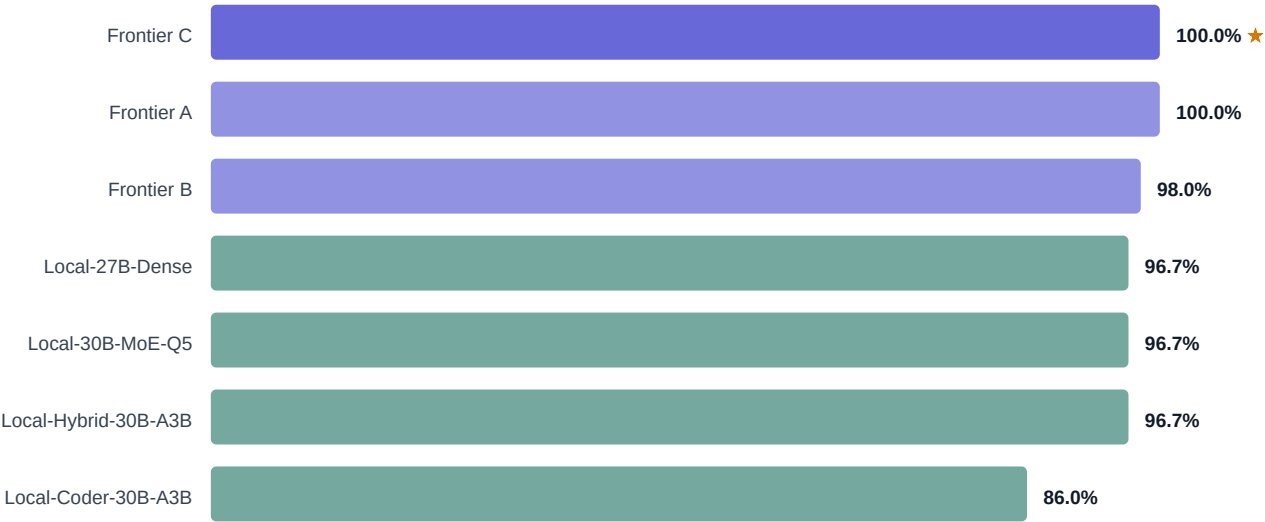
Quality index

Mean LLM-as-judge score across all 30 runs per model. Higher is better.



Pass rate

Mean per-test pass fraction of the hidden suites, with partial credit.



#	MODEL	CLASS	QUALITY	PASS RATE	MEDIAN LATENCY	SERVED AS
1	Frontier C undisclosed frontier model	FRONTIER	0.913	100.0%	40s	hosted API
2	Frontier B undisclosed frontier model	FRONTIER	0.911	98.0%	52s	hosted API
3	Frontier A undisclosed frontier model	FRONTIER	0.907	100.0%	56s	hosted API
4	Local-27B-Dense Dense 27B (MTP draft head)	LOCAL	0.844	96.7%	64s	Q4_K_M · 16 GB · 98,304 ctx
5	Local-30B-MoE-Q5 MoE 30B (A3B active)	LOCAL	0.844	96.7%	272s	Q5_K_M · 19 GB · 131,072 ctx
6	Local-Hybrid-30B-A3B Hybrid Mamba-2 + MoE 30B	LOCAL	0.800	96.7%	51s	IQ4_XS · 18 GB · 131,072 ctx
7	Local-Coder-30B-A3B MoE 30B coding-specialist	LOCAL	0.667	86.0%	144s	Q4_K_M · 18 GB · 65,536 ctx

The trade-off

Quality vs. latency

Up and to the left is better: higher judged quality, lower median time to a finished task. Log-scaled x-axis.



The headline result: the best local models land within **3.3 points of pass rate** and **0.07 of quality** of hosted frontier models, at comparable median latency — and one local model is faster than two of the three frontier endpoints. The gap that remains is concentrated in judged craft, not in whether the task got done.

How the weights were fitted to the card

Every local model is quantized to sit inside 24 GB of VRAM *with its KV cache at the context length it is actually served at* — the quantization is chosen per model rather than applied uniformly, and context is capped where the KV cache would otherwise crowd the weights off the card. Models are hot-swapped one at a time; only one is resident at any moment.

MODEL	NETDEVOPS	AIOPS	HPC SCHEDULING	INFERENCE SERVING	RAG PIPELINES	TOOL-CALLING LOOPS
Frontier C	100%	100%	100%	100%	100%	100%
Frontier B	100%	100%	100%	100%	70%	100%
Frontier A	100%	100%	100%	100%	100%	100%
Local-27B-Dense	100%	100%	50%	100%	100%	100%
Local-30B-MoE-Q5	100%	100%	50%	100%	100%	100%
Local-Hybrid-30B-A3B	100%	100%	50%	100%	100%	100%
Local-Coder-30B-A3B	75%	100%	100%	100%	90%	100%

Pass rate by domain, pooled across both harnesses. Dashes mark dimensions with no task in that tier for this model.

Same model, two harnesses

Because each model was driven by two different agent harnesses over identical tasks, the harness itself becomes measurable. It matters more than expected: the scaffolding around a model moves pass rate by up to 15 points on the same weights.

Same model, two agent harnesses

Pass rate for each model driven by each harness. Identical tasks, identical model, identical grading.

Claude Code Prime Agent



Disclaimer. This is not an official benchmark result for any model, vendor, product or organisation, and it is not endorsed by any of them. These are the numbers I measured on my own hardware, with my own task battery, my own quantizations and my own serving configuration, on a single day. Sample sizes are small (15 tasks per model per harness), the grading judge is itself a language model, and hosted frontier models are anonymised and served through an access path whose configuration I do not control. Results are not directly comparable to any published leaderboard and should not be read as a general claim about any model's capability. Treat this as one operator's engineering log, not as evidence.

Self-hosted models served locally on a single NVIDIA RTX 3090 (24 GB) over Thunderbolt 4. Task battery and runner published for reproduction. Frontier models anonymised by design.